# Simulation Object Configuration Module (SOCM)

**OOF™ Origin Open Foundation™**

Independent Methodological Authority

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## Governed Space

## Simulation Object Configuration

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## Canonical Definition

Simulation Object Configuration Module (SOCM) defines the governance conditions under which the specific configuration of a Simulation Object is identified, bounded, recorded, differentiated, changed, and maintained as a traceable simulation-governance state.

SOCM ensures that simulation activity remains bound not merely to the correct Simulation Object and version, but to the specific arrangement of components, settings, parameters, options, dependencies, interfaces, capabilities, and operational features that materially define the object state represented by the simulation.

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## Operational Role

SOCM serves as the fourth internal governance space of the Simulation Object Standard (SOS).

The preceding modules establish:

SOIM — What exactly is the Simulation Object?

SOBM — Where does the Simulation Object begin and end?

SOVM — Which materially distinguishable version of that object is being simulated?

SOCM asks:

What exact configuration of that Simulation Object Version is represented by the simulation?

The governed progression is:

Identified Object → Bounded Object → Object Version → Configuration Basis → Configuration Identification → Material Configuration State → Configuration Change → Simulation Binding → Governed Simulation Object Configuration

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## Module Operational Space

SOCM governs: Simulation Object configuration identity, configuration basis, configuration parameters, configuration components, configuration settings, configuration options, configuration dependencies, configuration interfaces, enabled capabilities, disabled capabilities, operational modes, feature states, component combinations, configuration baselines, configuration variants, configuration relationships, configuration changes, configuration materiality, temporary configurations, experimental configurations, configuration ambiguity, configuration conflict, configuration equivalence, configuration divergence, configuration applicability, configuration status, configuration history, configuration-to-simulation binding, configuration-to-output binding, cross-configuration applicability, and configuration traceability.

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## Module Function

A Simulation Object Version may exist in multiple configurations.

Those configurations may behave differently even though they belong to the same version.

An autonomous vehicle may operate with different sensor packages.

A robot may operate with different payloads.

An AI system may have different tools enabled.

A spacecraft may use different subsystem arrangements.

An industrial system may operate under different control settings.

A digital system may expose different capabilities depending upon deployment configuration.

Therefore:

Object Identity + Version Identity alone may be insufficient to establish what was actually simulated.

SOCM prevents: undefined configurations, silent configuration substitution, materially different configurations being treated as equivalent, configuration-dependent results being represented as version-wide results, configuration changes being hidden, outputs being detached from the configuration that produced them, temporary settings being mistaken for baseline configuration, and simulation conclusions being transferred between configurations without a governed basis.

## Simulation Object Configuration

A Simulation Object Configuration is the defined arrangement of simulation-relevant characteristics within a Simulation Object Version at a particular governed state.

A configuration may include: installed components, activated components, disabled components, software settings, control parameters, operational modes, feature flags, interfaces, connected subsystems, dependency states, tool permissions, sensor arrangements, actuator arrangements, hardware options, safety mechanisms, or other simulation-relevant object characteristics.

## Configuration Basis

The Configuration Basis establishes which object characteristics must be known to distinguish one simulation-relevant configuration from another.

The Configuration Basis should focus on characteristics capable of materially affecting: object behavior, available capabilities, interactions, assumptions, scenario response, simulation execution, outputs, reality correlation, or interpretation of simulation results.

## Configuration Identity

Where configuration differences can materially affect simulation outcomes, the represented configuration should possess sufficient identity to distinguish it from other configurations.

Conceptually:

## Simulation Object → Version → Configuration

For example:

## Robot R1 → Version 4.2 → Configuration C7

The configuration identifier may be internally assigned or derived from another controlled configuration system.

SIMULOS® does not require a particular naming convention.

## Configuration Baseline

A Simulation Object Configuration Baseline establishes the reference configuration represented by the simulation.

The baseline may identify: Simulation Object, object version, configuration identifier, material components, material settings, enabled capabilities, disabled capabilities, operational mode, material dependencies, relevant interfaces, effective state, and known configuration limitations.

## Material Configuration Element

A Material Configuration Element is any configurable characteristic capable of materially affecting the simulation or interpretation of its results.

Materiality is simulation-relative.

A setting irrelevant to one simulation may be critical to another.

Therefore:

Configuration Materiality must be evaluated relative to the Simulation Purpose and the behavior being represented.

## Configuration Completeness

Configuration Completeness asks:

Are all materially relevant configurable characteristics sufficiently identified to know which object state was actually represented?

Configuration completeness does not require documentation of every technically possible parameter.

It requires sufficient representation of the configuration characteristics material to the governed simulation.

## Configuration Variants

One Simulation Object Version may contain multiple valid configurations.

For example:

Version 5.0

may exist as:

## Configuration A — Standard

## Configuration B — Extended Capability

## Configuration C — Restricted Safety Mode

## Configuration D — Experimental

These remain configurations of the same version unless the underlying change satisfies the governed conditions for a new version.

## Parallel Configurations

Multiple configurations may exist simultaneously.

SOCM preserves their distinction where simulation results could differ materially between them.

Parallel configurations must not be collapsed into one generic object state merely because they share the same version.

## Temporary Configuration

A Simulation Object may temporarily enter a configuration used only for: testing, maintenance, emergency operation, degraded operation, experimentation, calibration, or another temporary purpose.

Where that configuration is simulated, its temporary nature should remain explicit.

## Experimental Configuration

An experimental configuration should remain distinguishable from the approved or operational baseline configuration.

Simulation findings from an experimental configuration must not silently become findings about the operational configuration.

## Configuration Change

A Configuration Change occurs when one or more configuration characteristics change while the underlying Simulation Object identity remains continuous.

Conceptually:

Version V1 / Configuration A → Configuration Change → Version V1 / Configuration B

A configuration change may or may not create a new Simulation Object Version.

## Configuration Change Materiality

A configuration change is material where it can materially affect: simulated behavior, available capabilities, scenario response, system interaction, expected outputs, reality representation, or interpretation of simulation results.

Material configuration changes should remain explicitly traceable.

## Configuration Change vs Version Change

SIMULOS® distinguishes:

## Configuration Change ≠ Version Change

A configuration change modifies the arrangement or state of configurable characteristics.

A version change establishes a materially distinguishable object generation or state requiring separate version identity.

For example:

## Version 3.0 / Sensor Package A

and:

## Version 3.0 / Sensor Package B

may represent two configurations of the same version.

A fundamental redesign of the sensing architecture may instead require:

Version 4.0

SOVM governs version identity.

SOCM governs configuration identity.

## Configuration Change Escalation

A configuration change may become sufficiently significant that the existing version classification should be reconsidered.

SOCM should therefore be capable of generating a:

## Version Reassessment Signal

where configuration change appears to exceed the current version boundary.

SOVM determines the resulting version consequence.

## Configuration-to-Simulation Binding

Every governed simulation should be capable of answering:

Which configuration of the Simulation Object Version was represented?

The relationship becomes:

## Simulation Object → Version → Configuration → Simulation Activity

This binding prevents simulation results from becoming detached from the actual configured state represented during simulation.

## Configuration-to-Output Binding

Where outputs depend materially upon configuration, the relationship should remain reconstructable:

## Object Configuration → Simulation Execution → Simulation Output

SOCM governs the configuration relationship.

It does not govern the substantive Simulation Output.

## No Silent Configuration Substitution Rule

SIMULOS® establishes:

A materially different Simulation Object Configuration must not replace the configuration represented by a simulation without explicit governance of the change.

If a simulation begins against Configuration A and execution materially shifts to Configuration B, the change must remain visible.

## No Configuration-Wide Assumption Rule

SIMULOS® establishes:

Results established for one configuration must not automatically be represented as applicable to every configuration of the same Simulation Object Version.

Shared version identity does not establish configuration equivalence.

## Cross-Configuration Applicability

A result established for Configuration A may remain applicable to Configuration B where the changed configuration characteristics do not materially affect the simulation question.

Possible applicability states may include:

## Fully Applicable

## Conditionally Applicable

## Partially Applicable

## Not Applicable

or:

## Applicability Undetermined

The basis should remain explicit.

## Configuration Equivalence

Two configurations may be treated as simulation-equivalent for a specific simulation purpose where their differences do not materially affect the characteristics being simulated.

Therefore:

## Technical Configuration Difference ≠ Automatic Simulation Difference

but equally:

## Shared Version ≠ Automatic Simulation Equivalence Configuration Divergence

Configuration Divergence exists where configurations differ sufficiently that simulation findings cannot legitimately be transferred between them without additional governance.

Divergence may result from differences in: components, capabilities, permissions, control settings, operational modes, dependencies, interfaces, safety mechanisms, or other material configuration elements.

## Configuration Ambiguity

Configuration Ambiguity exists where the exact configuration represented by the simulation cannot be established with sufficient confidence.

Material ambiguity should remain visible.

It must not be silently converted into an assumed configuration.

## Configuration Conflict

A Configuration Conflict exists where credible records indicate materially different configurations for the same Simulation Object during the same relevant simulation state.

The conflict should be resolved or explicitly preserved before configuration-dependent conclusions are relied upon.

## Configuration Status

A configuration may have a governed status such as:

Baseline

Operational

Alternative

Temporary

## Experimental

Degraded

Emergency

Superseded

Retired

Ambiguous

or:

## Undetermined

Configuration Status describes the relationship of the configuration to the Simulation Object.

It does not determine simulation validity.

## Configuration History

SOCM should preserve material configuration history where configuration changes affect interpretation of simulation activity.

Conceptually:

## Configuration A → Change → Configuration B → Change → Configuration C

This allows later reconstruction of which configuration existed at each relevant simulation point.

## Configuration Freeze

For some simulations, the represented configuration may need to remain fixed during execution.

A Configuration Freeze identifies the configuration state that must remain stable for the governed simulation activity.

Where a material change occurs during a freeze, the change should produce a governance signal rather than being silently incorporated.

## Dynamic Configuration

Some Simulation Objects legitimately change configuration during operation.

Examples include: adaptive robots, reconfigurable spacecraft, autonomous systems, modular industrial systems, dynamically provisioned computing systems, and AI systems with changing tool availability.

In such cases, SOCM should not artificially freeze the object.

Instead, it governs the allowed configuration state space and the transitions occurring within it.

## Configuration State Transition

For dynamically configurable objects, a simulation may intentionally represent:

## Configuration A → Configuration B → Configuration C

The transition itself may be part of the simulated behavior.

SOCM preserves which configuration states existed and when transitions occurred.

## Allowed Configuration State Space

Where dynamic configuration is expected, an Allowed Configuration State Space may define which configurations or configuration transitions belong to the intended simulation representation.

A transition outside that space may generate a configuration deviation signal.

## Configuration Deviation

A Configuration Deviation exists where the represented or executed configuration differs materially from the approved Simulation Object Configuration Basis.

Possible outcomes include:

## Accepted Deviation

## Conditional Deviation

## Material Deviation

## Configuration Conflict

or:

## Deviation Undetermined

SOCM records the deviation.

Downstream simulation governance determines its consequences.

## Configuration Dependency

Some configuration characteristics depend upon external or internal components.

SOCM records the configuration relationship where necessary to know the actual object state.

It does not assume governance ownership over the external dependency itself.

## Configuration vs Environment

SOCM governs characteristics belonging to the Simulation Object.

It does not govern the simulated environment surrounding the object.

For example:

## Robot Sensor Package

belongs to:

## Simulation Object Configuration

while:

## Warehouse Lighting Conditions

belong to the relevant simulation environment governance space.

The distinction preserves the object/environment boundary established by SOBM.

## Configuration vs Simulation Inputs

Configuration characteristics define the state of the object.

Simulation inputs are values, data, stimuli, or conditions supplied to simulation execution.

Therefore:

## Object Configuration ≠ Simulation Input

A parameter may appear technically similar in both contexts, but its governance role determines its architectural location.

## Configuration vs Scenario

A configuration describes:

what state the Simulation Object is in.

A scenario describes:

what governed situation the simulation subjects that object to.

SOCM does not govern scenario design.

## Configuration vs Reality Representation

SOCM identifies the actual object configuration claimed to be represented.

It does not determine whether the simulation model accurately represents that configuration.

That question belongs to the later Reality Representation governance space of SIMULOS®.

## Configuration vs Integrity

SOCM does not govern the general integrity of configuration records, components, or configuration processes.

That belongs to INTEGROS® where applicable.

SOCM governs specifically:

which configuration is relevant to the Simulation Object and how that configuration remains identifiable and traceable throughout simulation governance.

## Configuration vs Validation

SOCM does not determine whether a configuration has been validated for operational reliance.

That belongs to VALIDOS® where formal validation is required.

SOCM supplies the configuration identity and change information that validation may consume.

## Configuration Change Propagation

A material configuration change may affect downstream SIMULOS® governance, including: reality representation, assumptions, model and environment, inputs and data, scenario design, execution, outputs, reality correlation, simulation evidence, and operational reliance.

SOCM identifies the change and preserves its traceability.

It does not replace those downstream governance spaces.

## Simulation Object Configuration Record

SOCM should establish a persistent Simulation Object Configuration Record sufficient to reconstruct the configuration represented by the simulation.

The record may preserve: Simulation Object Identifier, Simulation Object Version Identifier, Simulation Object Configuration Identifier, Configuration Basis, Configuration Baseline, material components, material settings, material parameters, enabled capabilities, disabled capabilities, operational modes, dependencies, interfaces, configuration status, configuration variants, temporary states, experimental states, configuration changes, configuration deviations, configuration ambiguity, configuration conflicts, configuration equivalence, cross-configuration applicability, configuration history, simulation bindings, output references, and configuration traceability.

## Failure Conditions

SOCM governance failure may exist where: the Simulation Object Configuration cannot be identified, material configuration elements remain unknown, simulation activity cannot be bound to the configuration represented, materially different configurations are treated as equivalent without basis, configuration changes occur silently, configuration-dependent findings are represented as version-wide findings, temporary or experimental configurations are confused with operational configurations, configuration ambiguity is hidden, conflicting configuration records remain undisclosed, configuration transitions cannot be reconstructed, configuration changes are confused with version changes, object configuration is confused with environmental conditions, or downstream simulation conclusions cannot be traced to the configuration from which they originated.

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## Minimum Implementation Framework

### 1. Establish the Configuration Basis

Identify which configurable characteristics are material to the governed simulation. 2. Establish the Configuration Baseline

Record the material components, settings, capabilities, modes, interfaces, dependencies, and other configuration characteristics represented. 3. Bind Configuration to Object Version

Establish:

## Simulation Object → Version → Configuration

and preserve sufficient identity to distinguish materially different configurations. 4. Govern Configuration Change

Detect material configuration changes, preserve transitions and deviations, and trigger version reassessment where configuration change may exceed the current version boundary. 5. Preserve Simulation Traceability

Bind the relevant configuration to simulation activity and preserve sufficient history to determine which configuration produced or influenced downstream simulation outputs.

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## Governance Outputs

SOCM may produce: Simulation Object Configuration Identifier, Simulation Object Configuration Record, Configuration Basis Record, Configuration Baseline, Material Configuration Element Record, Configuration Completeness Assessment, Configuration Variant Record, Parallel Configuration Record, Temporary Configuration Record, Experimental Configuration Record, Configuration Change Record, Configuration Materiality Assessment, Configuration Transition Record, Configuration Freeze Record, Allowed

Configuration State Space, Configuration Deviation Record, Configuration Equivalence Assessment, Cross-Configuration Applicability Assessment, Configuration Divergence Record, Configuration Ambiguity Record, Configuration Conflict Record, Configuration Status, Configuration History Record, Version Reassessment Signal, Configuration-to-Simulation Binding, Configuration-to-Output Reference, and Simulation Object Configuration Traceability Map.

## Use Case 1 — Autonomous Robot Tool Configuration

### Scenario

An autonomous industrial robot operates under the same software and hardware version but may use different tools:

## Configuration A — Welding Tool

## Configuration B — Cutting Tool

## The control system version remains unchanged. Application

SOCM preserves the two configurations as distinct object states.

A simulation executed against the welding configuration remains bound to:

## Robot R1 → Version 5.2 → Welding Configuration

Simulation findings dependent upon tool geometry, payload, motion, or operational capability are not automatically transferred to the cutting configuration. Result

Shared version identity does not erase material configuration differences. Use Case 2 — Spacecraft Operational Configuration

### Scenario

A spacecraft operates in multiple governed configurations:

## Launch Configuration

## Orbital Configuration

## Docking Configuration

## The spacecraft remains the same object and version. Application

SOCM identifies each configuration and the permitted transitions between them.

A simulation evaluating docking behavior remains bound to the Docking Configuration rather than being represented generically as a simulation of the spacecraft in every operational state. Result

Simulation conclusions remain attached to the configuration actually represented.

## Architectural Position

Simulation Object Configuration Module (SOCM) is the fourth internal governance space of the Simulation Object Standard (SOS).

The internal progression is:

Simulation Object Identification → Simulation Object Boundary → Simulation Object Versioning → Simulation Object Configuration

SOIM establishes:

What is being simulated?

SOBM establishes:

Where does the object begin and end?

SOVM establishes:

Which version of the object is being simulated?

SOCM establishes:

What exact configuration of that version is being simulated?

The next internal governance space, SODM — Simulation Object Documentation Module, governs the persistent documentation necessary to preserve the complete governed Simulation Object definition.

## Foundational Principle

A simulation of the correct object and correct version may still represent the wrong operational reality if the configuration actually simulated is not the configuration to which its conclusions are applied.

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## Canonical Closing Statement

Simulation Object Configuration Module (SOCM) establishes the configuration-governance layer of the Simulation Object by ensuring that the specific arrangement of components, settings, parameters, capabilities, interfaces, dependencies, and operational modes represented by simulation remains explicitly identifiable and traceable. Through configuration baselines, materiality assessment, configuration variants, controlled change, dynamic configuration transitions, equivalence, divergence, deviation governance, cross-configuration applicability, and explicit simulation binding, SOCM prevents shared object or version identity from concealing materially different configured states. By preserving the relationship between Simulation Object identity, version, configuration, simulation activity, and downstream outputs, SOCM provides the fourth foundational object-governance mechanism of SIMULOS®.

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